MUSLIMHACKS 2026 · ELDERLY CARE · CONCORDIA UNIVERSITY

Amanah Care

The care record only your family can read.

Abdul Quayum Adekunle · safio

amanah = a trust you are responsible for

THE DIAGNOSIS

Ammi is 78. Three of her children share her care. This is the system.

Fatima · 07:52
did she take the morning ones?

Sister A · 08:10
yes. she ate half. who has her this afternoon?

Abdullah · 08:41
I thought Fatima. I have the pharmacy at 11 I think?

Fatima · 09:15
no that was last week. what did Dr Rahman say wednesday??

Sister A · 09:20
scroll up 🙃

- **No plan.** Things get missed, or done twice.
- **No handoff.** The next person does not know what happened.
- **No record of who did what.** One sibling carries it all.
- **No privacy.** Her health history sits with a chat company.

1 in 4 family caregivers say they struggle to coordinate care. Up from 1 in 5 five years earlier. AARP / NAC, Caregiving in the U.S. 2020

CARE IS ALREADY HAPPENING. IT IS JUST NOT TRACKED.

1 in 4

American Muslims care for an older adult, versus 16% of the general public.

ISPU, Support and Sacrifice, 2026

42%

of Canadians 15 and over gave unpaid care in 2022. 13.4 million people.

Statistics Canada, CSS 2022

49%

of family caregivers of people with dementia report burden; 31% depression.

Collins & Kishita, 2020, meta-analysis

What caregivers actually do, and where it lives in Amanah Care

TASK, SHARE OF CAREGIVERS	SOURCE	IN THE APP
Transportation, 72%	Statistics Canada 2018	**Pickups and drop-offs in the routine, with who is on it**
Scheduling and coordinating appointments, 40%	Statistics Canada 2018	**Appointments in the routine, next visit on the dashboard**
Medical and nursing tasks, 58%	AARP / NAC 2020	**Meds logged against the plan, 2 of 3 today**
Personal care, 60%	AARP / NAC 2020	**Modesty and caregiver-gender preference on every card**
Coordinating between caregivers, hard for 26%	AARP / NAC 2020	**Handoff, alerts, who carried the week**

A private care record. The plan, what was done, the handoff, who carried the week.

Routine	Ammi's plan, entered once. Every day becomes a checklist measured against it.
Log	Two taps: category, what happened. Meds 2 of 3, meals 2 of 3, prayers 3 of 5.
Hand off	"What happened" written from the log, "what is next" written from the routine. You type nothing.
Alerts	Each person picks what they want to hear about. Fatima: meds and appointments. Abdullah: pickups.
Ammi's screen	Her own view in big type: who is with her, what is coming, her prayers, who can read her record.
Family key	Made on the phone, shared by QR in person. The server stores scrambled text. Not even we can read it.

LIVE DEMO · THREE PHONES, ONE FAMILY

1. **Sister A** opens Home: Ammi's day, 10 of 12 done, who has Ammi, next pickup Sunday 11:00, Abdullah.
2. She taps **Done** on dinner. The tiles move.
3. She hands off to **Fatima**. Summary and "what is next" are already written. Send.
4. **Fatima's phone**: alert badge, the card with Ammi's preferences on top. Accept.
5. **Ammi's screen** now says "Fatima is looking after you today", in big type.
6. We open the **database**. Every row is scrambled text.

Watch for

- Nothing typed except one line.
- Three real sessions, one real backend, seconds apart.
- Readable on the phones, unreadable on the server.

HOW IT WORKS, AND WHY YOU CAN TRUST IT

1
Phone
Encrypts with the family key. AES-GCM, fresh IV each time.

READABLE

2
API
Checks membership, appends. Holds no key.

SEALED

3
Redis Streams
Append-only event log. Replayable.

SEALED

4
Projector + notifier
Two consumer groups: read models, alerts.

SEALED

5
Postgres
7 tables + 1 view. Metadata and ciphertext only.

SEALED

6
Family phones
Decrypt, compute the dashboard locally.

READABLE

Every action is an event

Dashboard, routine, record, alerts were added without touching the write path. They are projections over one log.

Zero knowledge, tested

A test writes "Ammi felt dizzy" and proves the sentence appears in no row, no stream entry, no API response. No LLM anywhere in the loop.

87 tests

99.8% line coverage on api, projector, notifier, crypto. Six containers, one command. One-process fallback.

Others built reminders, task lists, or a chatbot. We built the record.

- **Plan versus actual.** Adherence, not alarms. The handoff writes itself from the record.
- **Cryptographic privacy.** The server cannot read the data, and a test proves it. Not a policy.
- **The elder is in it.** Her own screen. Her preferences on every card. She sees who can read her record.
- **Real architecture.** Event sourced, two consumer groups, replayable, tested.
- **Grounded in the data.** Categories match what caregivers report doing.

Honest scope

REAL create, join, login, log, routine, handoff, alerts, dashboard, record, elder view, encryption, tests

SIMPLIFIED polling not push · demo default password 333 · one stream for all families · support workers share the family key

NOT BUILT elder consent filter (needs a scoped key) · key rotation · reminders · Urdu and Arabic · native apps · AI (edge only, roadmap)

WHO PAYS, WHAT IS NEXT, WHAT WE ASK

Who pays

Free for families. Paid per elder per month by home-care agencies and community programs that need handoff logs and scoped access. Ciphertext-only storage keeps hosting and compliance small.

Next

Monday: five Montreal families, one month pilot. Then a scoped key for support workers, push, key rotation, Urdu and Arabic on Ammi's screen. AI only on the phone, opt in.

We ask

Introductions to community care programs, five pilot families, a designer for the elder screen.

Care was already happening. Now it has a record. And only the family can read it.

BACKUP

Judge questions, answered

Technical · Impact and users · Demo and business · Team, process, data and privacy · Future and closing · Sources

Delivery format?	Web app, phone first, any browser. Two windows on stage stand in for two phones. Native mobile is roadmap.
Technologies?	Node/Express API, Redis Streams event log, Postgres read models, vanilla JS + WebCrypto AES-GCM, nginx, Docker Compose. Node's built-in test runner. Vendored QR library. No frameworks, no CDNs, so the demo needs no internet.
Behind the scenes?	Phone encrypts → POST /events → membership check → XADD to the stream → projector consumer group → Postgres events + handoffs + workload view; a second consumer group, the notifier, writes alerts per subscriber → phones poll and decrypt. The dashboard is computed on the phone.
Built from scratch here?	Spec, requirements and two audit rounds were planned ahead. All code was written during the hackathon window. QR generation is a vendored library.
Tradeoffs?	Polling over push, a demo default password over real credential policy, one stream over per-family streams, consent filter deferred. Each one on the honesty slide with the reason.

Target user?	Adult children sharing care of an elderly parent. The elder is the subject and owner of the preferences, not the app user. Support workers join with their own role.
Discovery and adoption?	Family to family, by QR, in person, by design. Community care programs and masjid welfare committees as the channel. Agencies bring their families.
How many benefit?	1 in 4 American Muslims already care for an older adult (ISPU 2026). Muslims are 8.7% of Greater Montreal (Census 2021). Start with Montreal families, then any family with a shared care load.
Impact if fully developed?	Fewer missed meds and appointments, a fairer split of the load, and less of the strain ISPU documents. A family-owned health record the family never had.
Success measure?	Plan adherence, handoff acceptance time, active caregivers per family, weekly return rate.

APPENDIX · DEMO, FUNCTIONALITY, BUSINESS

Fully functional or mocked?	The core loop is real end to end: encrypt, stream, project, store, decrypt. Login is real, with a demo default password. Notifications are polling. The consent filter is not built. Nothing on stage is faked.
Most important feature?	The handoff, composed from the record: what happened from the log, what is next from the routine, the elder's preferences on top.
Hardest part?	Keeping the server useful while it can read nothing: deciding which fields stay clear for routing and counting, and building a whole dashboard from client-side decryption.
Business model?	Free for families. Paid per elder per month by home-care agencies and community programs that need handoff logs and scoped access.
Scale beyond the prototype?	Per-family streams, push, scoped keys, password policy, offline. Ciphertext-only storage keeps compliance and hosting light.
Who pays?	Agencies and community organisations. Families optionally for extras like key escrow between two relatives.
What would it take to make it a real product?	Password policy and key recovery, native apps, scoped support keys, push, offline, accessibility, Urdu and Arabic.

How did you divide the work?	Abdul: architecture, event model, crypto, api and projector, dashboard, tests. safio: in-app invite flow, interface cleanup, the flow view that traces a write hop by hop.
What did you learn?	Cut scope early and say so. One workflow done well beats five half done. Privacy shapes every feature, including which ones you cannot build yet.
Differently next time?	Tests from hour one. Push notifications budgeted from the start.
How did you prioritise?	Requirement register of 46 items, two audit rounds on feasibility in 24 hours, then "cut core": the handoff loop first, everything else labelled roadmap.
What data?	Care events, handoff summaries, preferences, routine, member names, all encrypted. Clear: event type, category, member id, timestamps.
Ethical risks?	Support workers currently hold the same key, so hiding categories is UI only. The password-locked key copy is only as strong as the password, and the demo default is 333. Lost key means lost data until rotation ships. No diagnosis, dosage or triage anywhere, and the metrics are counts, never judgments.
Depends on an LLM?	No. By design the server cannot run one on this data.

Next if more time?	Urdu and Arabic for the elder view, scoped support key, push, key rotation, password policy and recovery.
Continue after today?	Yes. Five-family pilot in Montreal next week.
First step after today?	Commit and publish the repo, deploy on one small VM, recruit the pilot families through community care contacts.
Different from existing options?	Task apps organise chores, med apps remind, marketplaces find carers. None connect plan, actuals, handoff and load with the elder's preferences, and none let only the family read the data.
Help needed?	Introductions to community care programs and agencies, pilot families, and a designer for the elder view.
Remember most?	The server cannot read it. Open the database and see for yourself.
Why should this win?	It solves the coordination problem the track asked for, with the elder's preferences built in, privacy that is cryptographic rather than a promise, an honest scope, and a test suite behind it.

- ISPU, *Support and Sacrifice: Muslims Who Care for Older Adults Face Emotional Strain*, 2026, drawing on American Muslim Poll 2025. ispu.org/reports-and-analysis/support-and-sacrifice/
- ISPU, *Sandwiched: Muslims Are Most Likely to Be Caring for Aging Parents and Have School-Aged Children*. ispu.org/muslim-families/
- Statistics Canada, *More than half of women provide care to children and care-dependent adults in Canada, 2022*, The Daily, 8 Nov 2022. www150.statcan.gc.ca/n1/daily-quotidien/221108/dq221108b-eng.htm
- Statistics Canada, *Study: "Sandwiched" between multiple unpaid caregiving responsibilities*, The Daily, 2 Apr 2024. www150.statcan.gc.ca/n1/daily-quotidien/240402/dq240402d-eng.htm
- Collins, R. N. and Kishita, N., *Prevalence of depression and burden among informal care-givers of people with dementia: a meta-analysis*, Ageing & Society, 2020.
- MuslimHacks 2026 Challenge Guide, Elderly Care (research and fact sheets by Process Plus). References EC1 to EC6 therein, including Law 25, PHIPA, PIPEDA and Attum et al., *Cultural Competence in the Care of Muslim Patients and Their Families*, StatPearls.
- Statistics Canada, Census 2021 religion release: Muslims 4.9% of Canada, 8.7% of Greater Montreal (cited in the Challenge Guide).
- Chrome for Developers, *Built-in AI*: Prompt, Summarizer and Translator APIs on Gemini Nano. developer.chrome.com/docs/ai/built-in